LAB3: PLANAR MOBILE PHONE ANTENNA FOR AUTOMOTIVEAPPLICATIONS. BENCHMARKING

1. Objectives and preliminary considerations

The analysis of a mobile phone antenna which can be found in the automotive sector will be carried out within this laboratory session. The sample is a shark fin antenna commonly located in the rear part of the roof of the car.

The main goal of this lab is to understand the functionality of a real antenna sample as well as to get an idea of the production process, materials, costs, patents, specifications, etc. of some antenna parts. All this process is called as a benchmarking.

During the laboratory session, the student will carry out the benchmarking of the antenna with special focus on the simulation. For that reason, the real sample should be well recognized and understood to create a reliable CADFEKO model. The use of a ruler is needed to obtain the dimensions of the antenna.

The analysis will be performed in several steps to understand how each antenna part or element acts in the antenna performance. Step by step, the simulation model will represent more accurately the real antenna and the measurement results.

The 19th of April, there will be an introductory session where you will receive an explanation about the benchmarking process and most common antennas that nowadays are used in the automotive sector. The lab session will take place the 25th of April from 17 to 20 h at lab Q6-2008.

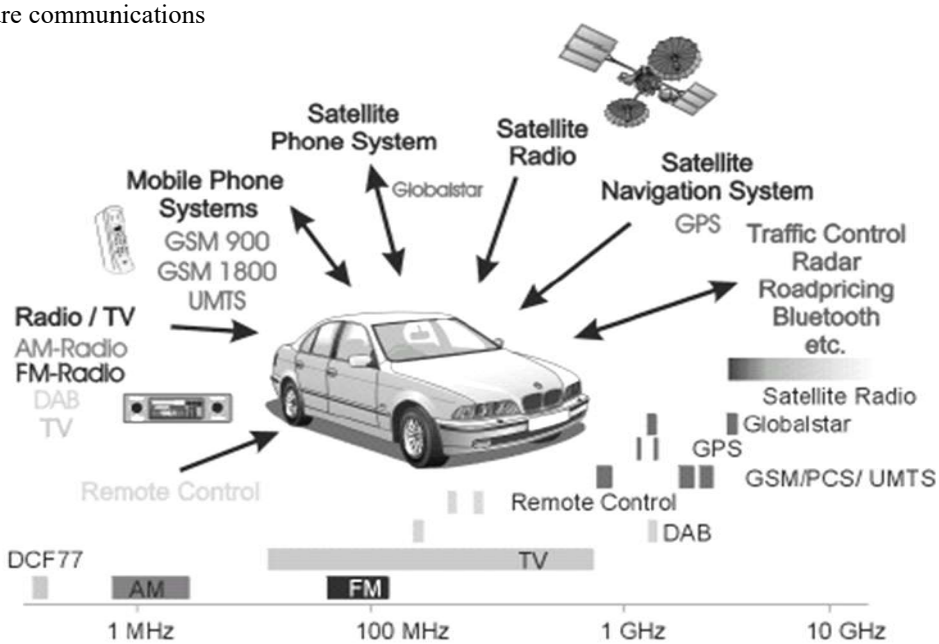
As a reminder, it is also recommended to read the last lab report and the document GetStarted.pdf from FEKO.

The deadline for delivering the report is the 11thof May. Please write your names in the file name.

2. Automotive antennas and measurements

2.1. Introduction

a) Today and near future communications
b) Integrated antennas
c) Planar structures

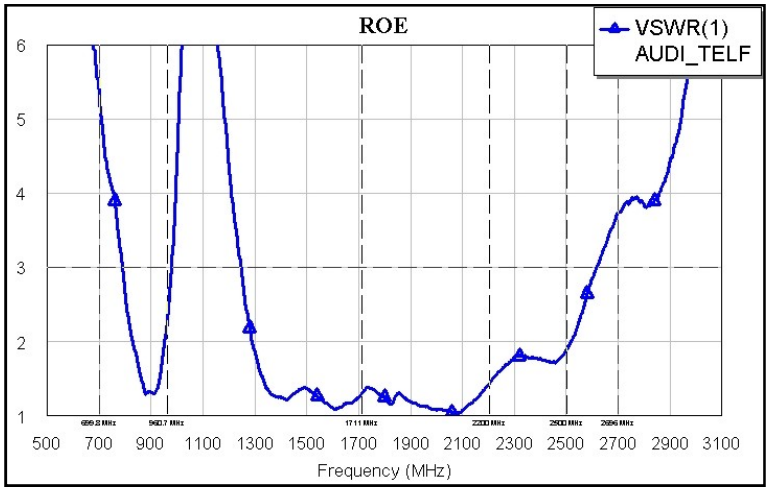


2.2. Antenna examples

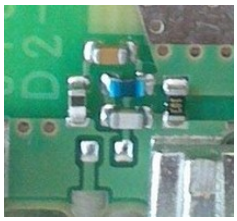
a) Antenna identification: services, typology, polarization, frequencies, technology, …
b) Shape analysis.
c) Mechanical parts and mounting process analysis: fixing system, antenna-PCB connection, coaxial soldering, waterproof, …

2.3. Antenna matching measurement

A real antenna matching measurement tells us about the services or the operating frequencies. The measurement is easily done and is useful to compare with simulation results.



2.4. Impedance matching network

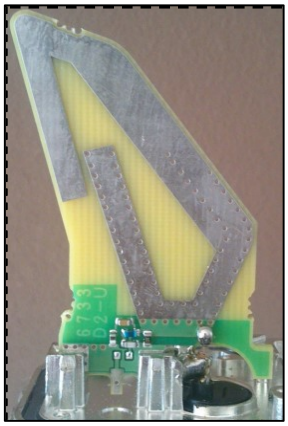


Wideband, multiband and miniaturized antennas commonly use a matching network to fulfil the customer spec about matching level. Up to 3GHz the matching network can be done with lumped components like SMD technology. At higher frequencies distributed transmission lines are used.The matching network can be also taken into consideration adding a [S] parameter box of the network in the simulation.
Other functionalities are antenna connection diagnosis, increase isolation, V+ feeding splitter...

2.5. Getting ready to build a FEKO model

Use a ruler to measure the antenna dimensions that you consider relevant to create a CADFEKO model. Keep in mind that the simulation result can depend on how you build the model.

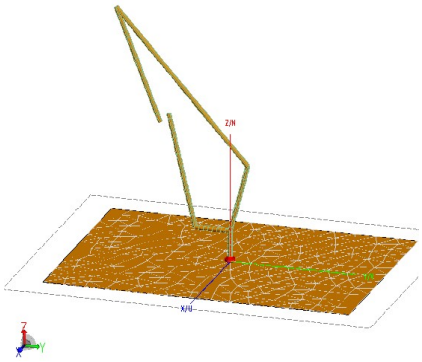
Real picture dimension 5.5 cm x 3.65 cm



3. Simulation

3.1. Wire model

Create a 3D model of the antenna with wires using the points measured in the real sample.



Create a ground plane with dimensions similar to the sample:
(12 cm x 6 cm).

Create a port and the voltage source on it.
Define the frequency points of simulation.
Mesh the structure with a meshing dimension according to the simulation frequency and the antenna structure shape.
Launch the simulation and analyse the parameters you consider as most important to understand how the antenna works.

Q1. Show the draft you use to define the geometry of the antenna with its dimensions. Explain the method used to define the linear approximation of your wire model geometry.

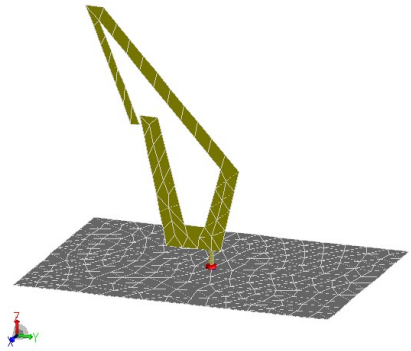
Q2. Plot the antenna impedance (real & imag), antenna matching and find the operating bands of the antenna. Comment the result. Is this antenna well matched to the telephony frequency bands?

Q3. Request the current distribution for the low and high frequency bands, at the frequencies where the antenna is better matched. Show the radiation pattern for both bands. Comment the result and explain which part of the antenna is the main contributor in the creation of each of the operating bands.

Note: Use an appropriate scale for accurate representation of currents distribution. For this exercise, I recommend to use linear scale for the currents plot

3.2. Planar model

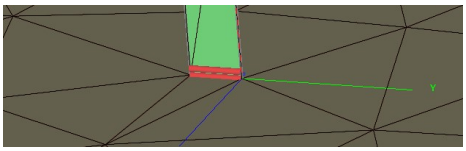
Replace the wire model with a surface model built with a polygon. Try to represent a 3D model similar to the real sample.



Define the conductor material with the following parameters:
Copper conductivity: $5.7.10^{7}$ S/m^{-1}
Aluminium conductivity: $3.8.10^{7}$ S/m^{-1}

Use Cooper for the antenna pattern and Aluminium for the ground plane. Use planar metal sheets with thickness 35µm

For feeding of the antenna use an edge port instead of wire port (you will need to replace the wire used to connect the antenna with the ground plane by a metal strip (polygon) and define an edge port). See an example in the following picture.



Mesh the model, choose the mesh size according to the simulation frequency and the antenna shape.Launch the simulation and analyse the parameters you consider relevant to understand how the antenna works.

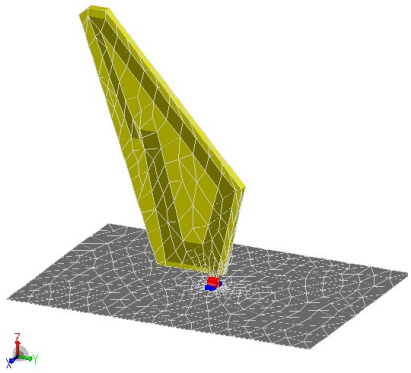
Compare the results with the previous simulation.

Q4. Show the generated model. Compare and comment the input impedance and the antenna matching of the antenna with the previous model in a cartesian plot and Smith chart.

Q5: Identify the 2 resonances in the smith chart and extract the impedance values at the resonance. Request the current distribution for the low and high frequency bands, at the frequencies where the antenna is better matched. Explain the results

3.3. Substrate effect

The substrate can be introduced using the polygon tool. Once the contour is done, with the sweep tool it is possible to create the volume of the antenna substrate.



Dielectric properties:

Relative permittivity: 4
Loss factor: 0.025
Thickness: 1.6 mm

Assign the media to the structure. In the geometry details sections, check that all the regions are faces are defined with the correct material. If not correct it

Mesh the model, choose the mesh size according to the simulation frequency and the antenna shape.
How many unknowns do you have?

Launch the simulation and analyse the parameters you considererrelevant to understand how the antenna works.

Q6. Show the generated model. Compare the results with the previous simulation (S11, VSWR, impedance,...).

Q7. Plot and comment the efficiency comparing the results with the previous models. Which are the efficiency levels in the bans of interest?

Note: If you are using a continuous interpolated frequency sweep, depending on the antenna dimensions you can obtain incorrect values in the efficiency (plot with peaks/artefacts that are clearly incorrect). If you have this problem in your results, then you need to modify the frequency sweep configuration to solve it (modify the converge accuracy in the continuous sweep, or use a linear discrete sweep)

Q8. Execute the simulation with different values of the Relative permittivity (2, 4, 6). Plot the S11 of the antenna for the 3 values of permittivity. Explain the main effect of the relative permittivity on the antenna behaviour

3.4. Tuning of the antenna

With the original permittivity value (4)., Modify the antenna geometry in order to adjust the operating bands of the antenna to the telephony frequency bands and to improve the impedance matching. Have a look at the currents on the surfaces to get better understanding about how the antenna works.

Q9. Explain the tuning process that you have done. Plot the last results of the antenna matching you have found during the tuning and compare it among the previous models all together in the same plot and comment the results. Which matching levels are you obtaining in the lower frequency bands of telephony? And in the higher frequency bands?

Q10: Add in the matching graph the result of the real measurement, use the file available in campus virtual (AUDI-TELF.s1p) and compare the results. Which would be your proposal for further improvements of the matching levels?